STAT 641 **Midterm Exam 1 Solution** **Ghosh**

1. (a) $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$, $\mu(A_n) = \infty$, $\bigcap_{n\geq 1} A_n = \emptyset$. So $\mu(\bigcap_{n\geq 1} A_n) \neq \lim_{n\to\infty} \mu(A_n)$. Hence m.c.f.a doesn't hold.

   (b) Take $\mu'(A) = \sum_{i\in A} \frac{1}{2^i}$. Then $\mu(\bigcap_{n\geq 1} A_n) = 0 = \lim_{n\to\infty} \frac{1}{2^{n-1}} = \lim_{n\to\infty} \mu(A_n)$. So m.c.f.a holds.

   (c) If $\exists\ n_0$ s.t $\mu(A_{n_0}) < \infty$, then m.c.f.a holds.

   (d) Always true.

2. Taking $F = \Omega \in \mathcal{F}$, we get $\Omega_0 = \Omega \cap \Omega_0 \in \mathcal{F}_0$.

   If $A = F \cap \Omega_0 \in \mathcal{F}_0$ for some $F \in \mathcal{F}$, then $\Omega_0 \setminus A = F^c \cap \Omega_0 \in \mathcal{F}_0$, since $F^c \in \mathcal{F}$ (note that $\Omega_0 \setminus A$ is the complement of A in Ω_0).

   If $\{A_i\} \subset \mathcal{F}_0$, then each $A_i = F_i \cap \Omega_0$, for some $\{F_i\} \subset \mathcal{F}$. In this case, $\cup_{i\geq 1} F_i \in \mathcal{F}$ and hence $\cup_{i\geq 1} A_i = (\cup_{i\geq 1} F_i) \cap \Omega_0 \in \mathcal{F}_0$. This proves that $\mathcal{F}_0$ is a σ-algebra of subsets of Ω_0.

3. (a) $f = 1_A$, where A is a non-Borel measurable set.

   (b) $g = 1_A - 1_{A^C} = \begin{cases} 1; & \text{on } A \\ -1; & \text{on } A^C \end{cases}$ is not measurable, but $|f| \equiv 1$ is Borel measurable (constant functions are always measurable).

5. (a) Note that for $B \in \mathcal{B}(\mathbb{R})$,

$$T^{-1}(B) = \begin{cases} \emptyset; & \text{if } 5 \notin B \text{ and } 0 \notin B \\ (-\infty, 0]; & \text{if } 5 \in B \text{ and } 0 \notin B \\ (0, \infty); & \text{if } 5 \notin B \text{ and } 0 \in B \\ \mathbb{R}; & \text{if } 5, 0 \in B \end{cases}$$

   Since $(-\infty, 0], (0, \infty) \notin \mathcal{F}_1$, then T is not $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ measurable.

   (b) Note that $\{\epsilon_i : i \geq 1\}$ are measurable, and for each $i \geq 1$, Y_i is created by adding a constant to ϵ_i and hence Y_i is measurable. For each $n \geq 1$, $\hat{\beta}_n$ is created by multiplication, component-wise addition and division(multiplication) by a constant of measurable functions $\{Y_i : i \geq 1\}$ and hence they are measuable too.